

Principles and Tools in Biological Data Analytics I
Exercise Sheet 2 — 26 September 2026

Problem 9

Write code that prints to the console elements of any given list present at odd index positions. Use as example input [10, 20, 30, 40, 50, 60, 70, 80, 90, 100].

Problem 10

Write code to print the following pattern for a maximum depth n to the console.

```
*
* *
* * *
* *
*
```

Problem 11

Write code to print the alternate numbers pattern for a given number of rows r . The alternate numbers pattern contains in each row only as many numbers n as the row number r_i itself.

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

Problem 12

Write code to calculate the factorial of a given number n .

NOTE: The factorial of n is denoted as $n!$ and is calculated as $n \times n - 1 \times n - 2 \times \dots \times 2 \times 1$.

Problem 13

Write code to display the entire Fibonacci sequence up to a given number n .

NOTE: The Fibonacci sequence F is a sequence of numbers where the current Fibonacci number F_i is the sum of the two preceeding Fibonacci numbers $F_{i-2} + F_{i-1}$

Problem 14

Write code that builds the complement sequence for a given DNA sequence. Print the result as follows:

```
5 '—ATCGTGAATCG—3 '
   |||||
3 '—TAGCACTTAGC—5 '
```

Problem 15

Building on Problem 14 write code that builds the reverse complement sequence for a given DNA sequence (i.e. both sequences should be $5' \rightarrow 3'$). Print the result as follows:

```
5 '—ATCGTGAATCG—3 '
5 '—CGATTACGAT—3 '
```

Problem 16

Checking for sequence similarity for short DNA/RNA sequences is oftentimes done using a Dot-Matrix. Given two sequences A and B implement a comparison algorithm that prints the sequence A as the first line in the output and the sequence B down on the left side. Fill the matrix whenever the $A[i] == B[j]$ with the symbol *. The output for the two example sequences A=GATTCGCTTAGT and B="CTGATTCCTTAGTCAG" should look like this:

```
GATTCGCTTAGT
C      *  *
T    **      **  *
G*      *      *
A  *              *
T    **      **  *
T    **      **  *
C      *  *
C      *  *
T    **      **  *
T    **      **  *
A  *              *
G*      *      *
T    **      **  *
C      *  *
A  *              *
G*      *      *
```